AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

الجامعة العربية المفتوحة
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 21 /Apr./2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 3 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which of the following activities for the development of good software system is concerned with controlling the cost of developing such systems:
   a. Quality management b. Project management c. Design d. Maintenance

2. The set of rules that define how a software development project should be carried out is knows as:
   a. Development process b. Deliverable c. Quality management d. Use Case development

3. Representing the marketing team of a supermarket as a class in a system is best categorized as which category
   a. Tangible Objects b. Roles c. Events d. Organizational Units

Part II – Short Answer Questions — 12 Marks

*Answer **all** the following short answer questions. Each question is worth 3 Marks.*

1. Why do legacy systems lack in flexibility?
2. State the inputs and outputs of the requirements engineering process.
3. For each of the following requirement, state if it is functional or non-functional and also specify its type.
   a- The application shall use the company's official brand colors, logo, and typography consistently across all screens
   b- Page load time shall not exceed 3 seconds on a standard broadband connection
   c- The system must allow administrators to approve or reject user account registrations
4. Briefly state what is meant by User-Centred design and which diagram is manly related to it.

Part III – Problem Solving Questions — 15 Marks

*Answer **all** the following problem-solving questions. Each question is worth 7.5 Marks.*

1. Develop a Use Case diagram for registering courses at a university with the following requirements:
   - A Faculty member can check student grades
   - A Faculty member can be either an Advisor or a Tutor
   - A Tutor can update student grades
   - An Advisor can view offered courses and register courses
   - A Student can view offered courses
   - All Faculty members must log in before doing any other task

Note: Make sure to use Generalization and Stereotype properly



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2. Develop a class diagram for a Virtual Learning Environment with the following requirements:
   a- Represent 3 main classes (User, Course, Assessment).
   b- Represent using generalization and additional classes how a User can be either a Member or a Visitor and how a Course can be either a FreeCourse or a PaidCourse.
   c- Use associations as you find suitable and make sure to use Aggregation/Composition at least once.
   d- Add the respective multiplicities and role names.

End of questions



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